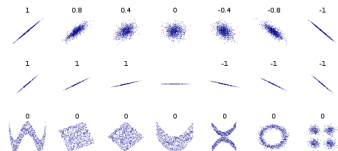
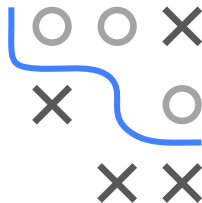


Introduction to Machine Learning

Feature Selection

Feature Selection: Filter Methods

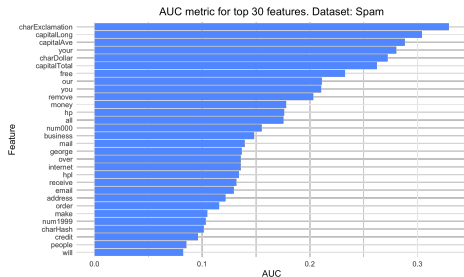
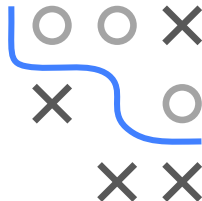


Learning goals

- Understand how filter methods work and how to apply them for feature selection.
- Know filter methods based on correlation, test statistics, and mutual information.

INTRODUCTION

- **Filter methods** construct a measure that quantifies the dependency between features and the target variable
- They yield a numerical score for each feature x_j , according to which we rank the features
- They are model-agnostic and can be applied generically



Exemplary filter score ranking for Spam data

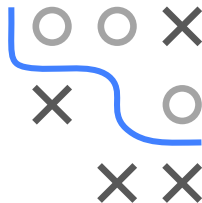
χ^2 -STATISTIC

- Test for independence between categorical x_j and cat. target y .
Numeric features or targets can be discretized.
- Hypotheses:
 $H_0 : p(x_j = m, y = k) = p(x_j = m) p(y = k) \forall m, k$
 $H_1 : \exists m, k : p(x_j = m, y = k) \neq p(x_j = m) p(y = k)$
- Calculate χ^2 -statistic for each feature-target combination:

$$\chi_j^2 = \sum_{m=1}^M \sum_{k=1}^K \left(\frac{e_{mk} - \tilde{e}_{mk}}{\tilde{e}_{mk}} \right)^2 \underset{\text{approx.}}{\overset{H_0}{\sim}} \chi^2((M-1)(K-1)),$$

where e_{mk} is observed relative frequency of pair (m, k) ,
 $\tilde{e}_{mk} = \frac{e_{m \cdot} \cdot e_{\cdot k}}{n}$ is expected relative frequency, and M, K are number of values x_j and y can take

- The larger χ_j^2 , the more dependent is the feature-target combination $\rightarrow$ higher relevance



PEARSON & SPEARMAN CORRELATION

Pearson correlation $r(x_j, y)$:

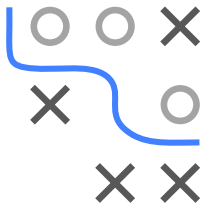
- For numeric features and targets only
- Measures linear dependency

$$\bullet \quad r(x_j, y) = \frac{\sum_{i=1}^n (x_j^{(i)} - \bar{x}_j)(y^{(i)} - \bar{y})}{\sqrt{\sum_{i=1}^n (x_j^{(i)} - \bar{x}_j)^2} \sqrt{\sum_{i=1}^n (y^{(i)} - \bar{y})^2}}, \quad -1 \leq r \leq 1$$

Spearman correlation $r_{SP}(x_j, y)$:

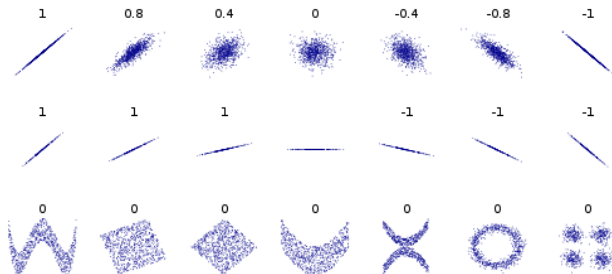
- For features and targets at least on ordinal scale
- Equivalent to Pearson correlation computed on ranks
- Assesses monotonicity of relationship

Use absolute values $|r(x_j, y)|$ for feature ranking:
higher score indicates a higher relevance



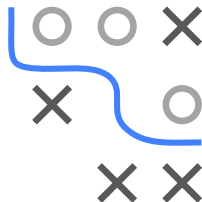
PEARSON & SPEARMAN CORRELATION

Only **linear** dependency structure, non-linear (non-monotonic) aspects are not captured:



Comparison of Pearson correlation for different dependency structures.

To assess strength of non-linear/non-monotonic dependencies, generalizations such as **distance correlation** can be used.



WELCH'S t-TEST

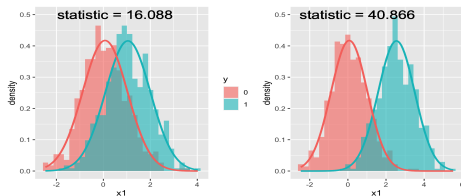
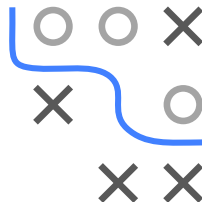
- For binary classification with $\mathcal{Y} = \{0, 1\}$ and numeric features
- Two-sample t-test for samples with unequal variances
- Hypotheses: $H_0: \mu_{j_0} = \mu_{j_1}$ vs. $H_1: \mu_{j_0} \neq \mu_{j_1}$

- Calculate Welch's t-statistic for every feature x_j

$$t_j = (\bar{x}_{j_0} - \bar{x}_{j_1}) / \sqrt{(S_{x_{j_0}}^2 / n_0 + S_{x_{j_1}}^2 / n_1)}$$

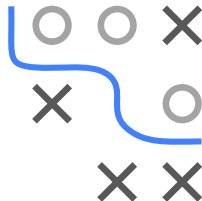
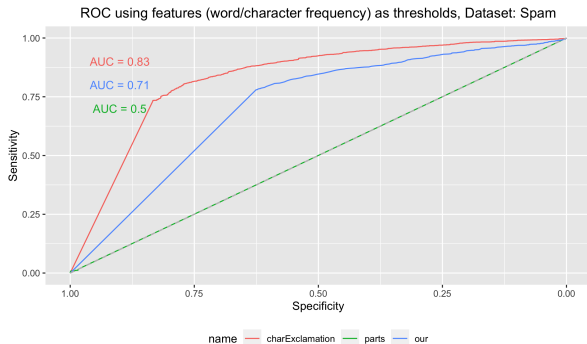
($\bar{x}_{j_y}$, $S_{x_{j_y}}^2$ and n_y are the sample mean, variance and sample size)

- Higher t-score indicates higher relevance



AUC/ROC

- For binary classification with $\mathcal{Y} = \{0, 1\}$ and numeric features
- Classify samples using single feature (with thresholds), compute AUC per feature as proxy for its ability to separate classes
- Features are then ranked; higher AUC scores $\rightarrow$ higher relevance.



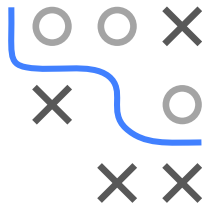
F-TEST

- For multiclass classification ($g \geq 2$) and numeric features
- Assesses whether the expected values of a feature x_j within the classes of the target differ from each other
- Hypotheses:
 $H_0 : \mu_{j_0} = \mu_{j_1} = \dots = \mu_{j_g}$ vs. $H_1 : \exists k, l : \mu_{j_k} \neq \mu_{j_l}$
- Calculate the F-statistic for each feature-target combination:

$$F = \frac{\text{between-group variability}}{\text{within-group variability}}$$
$$F = \frac{\sum_{k=1}^g n_k (\bar{x}_{j_k} - \bar{x}_j)^2 / (g - 1)}{\sum_{k=1}^g \sum_{i=1}^{n_k} (x_{j_k}^{(i)} - \bar{x}_{j_k})^2 / (n - g)}$$

where $\bar{x}_{j_k}$ is the sample mean of feature x_j where $y = k$ and $\bar{x}_j$ is the overall sample mean of feature x_j

- A higher F-score indicates higher relevance of the feature



MUTUAL INFORMATION (MI)

$$I(X; Y) = \mathbb{E}_{p(x,y)} \left[\log \frac{p(X, Y)}{p(X)p(Y)} \right]$$

- Each feature x_j is rated according to $I(x_j; y)$; this is sometimes called information gain
- MI measures the amount of "dependence" between RV by looking how different their joint dist. is from strict independence $p(X)p(Y)$.
- MI is zero iff $X \perp\!\!\!\perp Y$. On the other hand, if X is a deterministic function of Y or vice versa, MI becomes maximal
- Unlike correlation, MI is defined for both numeric and categorical variables and provides a more general measure of dependence
- To estimate MI: for discrete features, use observed frequencies; for continuous features, binning, kernel density estimation is used

